R. H. Indra

Machine Learning Engineer|Python|Scikit-Learn|TensorFlow|Deep Learning|Financial Data Analysis

Email: indradata96@gmail.com
Mobile: +91 88258 72155

Professional Summary:

Machine Learning Engineer with strong expertise in Python, SQL, and predictive modelling.Hands-on experience to build predictive models and data-driven solutions.Skilled in data preprocessing, feature engineering, model training, and model evaluation using Scikit-learn and TensorFlow. .Having professional experience working with large-scale financial transaction datasets at Barclays,enabling strong understanding of structured data, transaction patterns, and risk analysis.Developed multiple Machine Learning projects including fraud detection, credit risk,customer purchase prediction with a focus on solving real-world financial problems.Seeking an opportunity to grow and create impact as a Data Scientist or Machine Learning Engineer or Artificial Intelligence Engineer where I can continue my learning and contribute effectively with my technical skills for the development of the organization.

Professional Experience:

I am currently working in Barclays Shared Services Ltd during the period of October 14,2019-Present with 6 years and 6 months of experience.Analyzed large scale financial transaction datasets to identify anomalies and risk patterns.Performed data validation and reconciliation to maintain data integrity and accuracy.Worked with structured financial datasets including currency values, beneficiary data, and transaction attributes.Generated insights to improve payment processing efficiency and reduce transaction errors.Collaborated with cross functional teams to resolve errors in high volume data workflows.Applied data analysis techniques using Python libraries such as Pandas and NumPy for exploratory analysis.Hands-on experience building machine learning models on financial datasets including fraud detection,credit risk prediction.

Technical Skills:

Programming & Query Languages

- Python and SQL

Data Analysis & Visualization

- Pandas and NumPy
- Matplotlib and Seaborn

Machine Learning

- Scikit-learn,Supervised Learning (Regression, Classification)
- Unsupervised Learning (Clustering, PCA)
- Feature Engineering and Model Evaluation

Deep Learning

- TensorFlow / Keras
- ANN, CNN (Basic implementation)

Techniques: Feature Engineering, Model Evaluation

- Hyper parameter Tuning, Cross Validation

Tools & Platforms

- Jupyter Notebook, GitHub or Google Colab

Roles and Responsibilities:

- Processed and validated high-volume financial transaction data, ensuring accuracy and compliance with banking standards.
- Performed data validation, reconciliation, and quality checks on structured transaction datasets.
- Conducted Exploratory Data Analysis (EDA) using Python, Pandas, and NumPy to identify trends and patterns in structured financial data.
- Worked extensively with large structured financial datasets, strengthening data handling and preprocessing skills.
- Applied data cleaning and transformation techniques to ensure data consistency and reliability.
- Generated insights from transaction data to improve process efficiency and error reduction.
- Evaluated model performance using Accuracy, Precision, Recall, F1-Score, and ROC-AUC metrics to assess model effectiveness.
- Implemented data visualization techniques using Matplotlib and Seaborn to identify patterns in datasets.
- Developed and implemented Artificial Neural Network(ANN) for Classification, regression and Convolutional Neural Network (CNN) for Image recognition, object detection and Convolutional Neural Network for Image recognition, object detection

Project 1: Payment Fraud Detection

Client Name: Angel

My Role: Machine Learning Engineer

Project Duration: 4 months

Team Members: 4

Tools: Jupyter Notebook,Kaggle dataset

Technologies: Python,Pandas,Numpy,Scikit-learn,Matplotlib,XGBoost,Decision Tree,Random Forest

Project Description:

This project aims to identify suspicious financial transaction. Performed data preprocessing, feature engineering, and exploratory data analysis to understand transaction patterns. Implemented classification algorithms including Random Forest and XGBoost and evaluated model performance using precision, recall, F1-score, and ROC-AUC metrics to ensure effective fraud detection.

Roles and Responsibilities:

- Developed a machine learning model to detect fraudulent financial transactions using transaction datasets.
- Implemented classification models including Logistic Regression, Random Forest, and XGBoost.

- Performed data preprocessing, feature engineering, and exploratory data analysis (EDA) to identify patterns in financial transaction behavior.
- Evaluated performance using precision, recall, F1-score, and ROC-AUC metrics

Project 2: Credit Risk Prediction- Loan Default

Client Name: Austin

My Role: Machine Learning Engineer

Project Description:

This project aims to predict loan default risk using borrower financial and demographic data.Built a machine learning model to predict loan default risk using financial datasets.Built machine learning models for credit risk classification like Decision Tree.Evaluated model performance using classification metrics using precision, recall, F1-score, and ROC-AUC metrics to improve credit risk prediction accuracy.

Roles and Responsibilities:

- Built a machine learning model to predict loan default risk using financial and demographic data.
- Applied data preprocessing, feature engineering, and classification algorithms.
- Implemented credit risk prediction accuracy using classification metrics like Logistic Regression and Random Forest models.
- Performed feature engineering and cross validation for model evaluation.

Project 3: Sentiment Analysis - Natural Language Processing(NLP)

Project Description:

Developed a Sentiment Analysis system using Natural Language Processing (NLP) to classify text data as positive, negative, or neutral. Implemented text preprocessing techniques such as tokenization, stopword removal, and stemming/lemmatization, and trained machine learning/deep learning models to achieve accurate sentiment classification.

Roles and Responsibilities:

- Built an NLP model to classify text sentiment as positive, negative, or neutral.
- Performed text preprocessing including tokenization, stop word removal, lemmatization, and vectorization.
- Implemented machine learning models including Logistic Regression and Naive Bayes.
- Removing HTML tags and special characters, remove stopwords by using NTLK(Natural Languange Tool Kit)

Project 4 : Facial Age Recognition

Project Description:

This project aims to identify Facial Age Recognition is a computer vision–based AI system that automatically estimates a person's age from a facial image. The project uses deep learning techniques, particularly Convolutional Neural Networks to analyze facial features such as wrinkles, skin texture, and facial structure.The system is trained on large datasets of labeled face images with corresponding age

information. Once trained, it can predict the approximate age of a person from a new input image in real time.Technologies used: Python, TensorFlow, Keras, NumPy, Pandas, Jupyter Notebook

Roles and Responsibilities:

- Designed and developed a CNN-based model for age prediction using Python, TensorFlow, Keras.
- Preprocessed image data (face detection, resizing, normalization, augmentation) to improve model performance
- Models can be divided into three set of layers **(Input layer, Hidden layer, Output layer)**Input layer can be designed to make the number of layers into flatten shape.
- Activation function **'RELU'(Rectified Linear Unit)** can be helped to push the values from one neuron to another neuron.
- Compiling the training process and evaluated the model by using optimizer,loss and metrics function and specify the **Epochs** to iterate the model.

Project 5: CIFAR10 Classification-

Project Description:

This project aims to classify built in CIFAR10 datasets for analyzing both Artificial Neural Network (ANN) model and Convolution Neural Network (CNN) model. Developed a Convolution Neural Network (CNN) to classify images from the CIFAR-10 datasets containing 60,000 labeled images across 10 classes.Designed a deep learning architecture with convolution, max-pooling, and fully connected layers to reduce over fitting. Built a CNN-based deep learning model using Python and TensorFlow to classify images into 10 categories.

Appreciation and Achievements:

- Got The Best Project award for completing my final year project in Post Graduate education.
- Participated in college symposium and worked as a organizer to organize the program.
- Participated in social activities for planting trees with office colleagues .
- Achieved year end Best Performance award in my previous organization.
- Got Goody cupboard winners for February month in my latest organization.

Education:

- Completed Master of Business Administration from K.L.N college of Engineering, Madurai with First Class

Declaration:

I hereby declare that the information furnished above is true and correct to the best of my knowledge.

Yours Faithfully,

(R.H.INDRA)